

LECTURE 18: MATH1061/7861
TOPICS: SET THEORY DEFINITIONS, MORE
DEFINITIONS AND EXAMPLES OF SETS
DATE: THURSDAY APRIL 2, 2026

Learning objectives:

(1) Learn the notation for sets, such as:

$$x \in A, \quad B \subseteq A, \quad B \subset A, \quad A \cap B,$$

$$A \cup B, \quad A - B, \quad A^c, \quad \mathcal{P}(A), \quad A \times B.$$

(2) Understand the difference between $\in$ ('is an element of') and $\subseteq$ ('is a subset of').

(Deadline: Friday April 10th)
Anonymous survey link will be released
this afternoon. Results go to class reps.

Q1: Let

$$D = \{n \in \mathbb{Z} \mid n \equiv 2 \pmod{8}\}$$

E = the set of even integers

$$= \{\dots, -14, -6, 2, 10, 18, \dots\}$$

$$= \{\dots, -4, -2, 0, 2, 4, \dots\}$$

True or false: (Explain)

(a) $D \subseteq E$ True.

$$n \equiv 2 \pmod{8} \text{ means } n = 8k + 2$$

for some $k \in \mathbb{Z}$. Thus, $n = 2(4k + 1)$

So, n is even.

(b) $E \subseteq D$ False. Because $4 \in E$ but $4 \notin D$.

Q2: Determine the following sets:

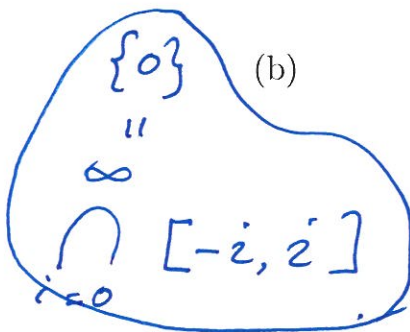
(a)

$$\bigcup_{i=0}^{\infty} \{i, -i\}$$

$\{i, -i\}$
Set with two elements

$$\begin{aligned} \bigcup_{i=1}^{\infty} \{i, -i\} &= \{-1, 1\} \cup \{-2, 2\} \cup \dots \\ &= \mathbb{Z} \end{aligned}$$

(b)



$$\bigcap_{i=0}^{\infty} [-i, i]$$

$$\bigcup_{i=0}^{\infty} [-i, i]$$

$[-i, i]$
 $= \{a \in \mathbb{R} \mid -i \leq a \leq i\}$

$$\begin{aligned} \bigcup_{i=1}^{\infty} [-i, i] &= [-1, 1] \cup [-2, 2] \cup \dots \\ &= \mathbb{R} \end{aligned}$$

(d) Determine $A_2 \times A_3$.

$$A_2 = \{1, 2\}$$

$$A_3 = \{1, 2, 3\}$$

$$A_2 \times A_3 = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3) \\ (2, 1), (2, 2), (2, 3) \end{array} \right\}$$

Is $A_2 \times A_3 = A_3 \times A_2$? **No.**

$$A_3 \times A_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$\mathcal{P}(A) = \text{set of subsets of } A.$

Q3: Let $A_n = \{1, 2, 3, \dots, n\}.$

(a) Write explicitly:

1

$$\mathcal{P}(A_0) = \{\emptyset\}$$

2

$$\mathcal{P}(A_1) = \{\emptyset, \{1\}\}$$

4

$$\mathcal{P}(A_2) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

8

$$\mathcal{P}(A_3) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

$$A_0 = \emptyset$$

$$A_1 = \{1\}$$

$$A_2 = \{1, 2\}$$

$$A_3 = \{1, 2, 3\}$$

$\vdots$

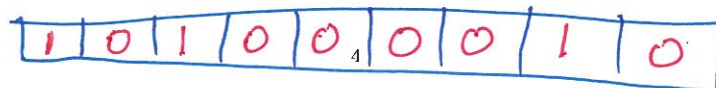
(b) Guess a formula for the number of elements in $\mathcal{P}(A_n).$ 2^n

Prove it!

(c) Challenge: Prove your guess is correct

One proof: Let $A \subseteq \{1, 2, 3, \dots, n\}.$

We build a string of length n
consisting of 0's and 1's using $A.$



Total # of strings is $2^n.$

$$\{1, 3, 8\} \cap A_9$$